

# Literature status: plasticity of $B_{\ell_1 \oplus_p \mathbb{R}}$

Run fa\_banach\_001 — lane 12

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**Status.** *Literature already answered: the  $\ell_1 \oplus_p \mathbb{R}$  subcase for every  $1 < p < \infty$ .* This packet records an exact later source; it is not a new proof.

## 1 Source question and special case

Haller, Leo, and Zavarzina’s *Two new examples of Banach spaces with a plastic unit ball* (arXiv:2111.10122; published 2022) asks in its introduction, PDF page 1, whether the unit ball of every Banach space is plastic. Here a metric space is plastic when every non-expansive bijection from the space onto itself is an isometry. The paper’s Theorem 3 proves the particular case

$$B_{\ell_1 \oplus_2 \mathbb{R}} \text{ is plastic.}$$

A natural direct attack is to replace the outer Euclidean norm by an arbitrary  $p$ -norm. The source proof does extend: strict convexity of  $\ell_p^2$  replaces the Euclidean antipodality calculation, and the scalar functions

$$A_p(t) = 1 - |t|^p + |1 - t|^p, \quad B_p(t) = 1 - |t|^p + |1 + t|^p$$

are respectively strictly decreasing and strictly increasing on  $[-1, 1]$ . The remaining layer-rigidity argument is unchanged after replacing squares by  $p$ th powers.

## 2 Exact later answer

Nikita Leo’s 2024 University of Tartu Master’s thesis *Banachi ruumi ühikkera plastilisus* (English title: *Plasticity of the Unit Ball of a Banach Space*) contains the exact result in Chapter 2. At the start of the chapter (printed page 14; PDF page 14), the thesis cites the Haller–Leo–Zavarzina paper, explicitly notes that the paper treats only  $p = 2$ , and says that the chapter generalizes it to arbitrary real  $p > 1$ . It then states:

**Theorem 2.1 (translated).** For every  $p \in (1, \infty)$ , the unit ball of  $\ell_1 \oplus_p \mathbb{R}$  is plastic.

The thesis supplies the proof through the rest of Chapter 2. Its English abstract on PDF page 2 also states the all- $p$  result. The supporting author therefore knew and stated that this was a generalization of the source paper; the match is explicit rather than agent-inferred.

### 3 Scope and search record

This settles the entire finite- $p$  family. The endpoint  $p = 1$  is isometric to  $\ell_1$  and is already classical. Neither the thesis identification nor this packet settles  $p = \infty$  or the source’s universal question for arbitrary Banach spaces.

The four run indexes and the local parsed arXiv corpus contained no duplicate. Bounded exact web searches for plastic  $p$ -sums first found only the source; a subsequent primary-source search located the 2024 University of Tartu thesis and its exact Chapter 2 statement. The official supporting PDF was inspected through DSpace, although direct bitstream transfer repeatedly stalled in the local environment. Stable repository links and exact locations are preserved in `supporting_thesis_metadata.md`.

### References

1. R. Haller, N. Leo, and O. Zavarzina, *Two new examples of Banach spaces with a plastic unit ball*, Acta Comment. Univ. Tartu. Math. 26 (2022), 89–101; arXiv:2111.10122.
2. N. Leo, *Banachi ruumi ühikera plastilisus* [*Plasticity of the Unit Ball of a Banach Space*], Master’s thesis, University of Tartu, 2024, Chapter 2, Theorem 2.1, <https://hdl.handle.net/10062/100461>.